

Valve Spring Finite Element Analysis

Drawing A177 053 05 00 | Intake Valve Spring (Beehive)

Spring Geometry

Free length:	46.1 mm
Wire cross-section:	2.92 × 3.66 mm (oval)
Total coils:	8.6 (1.25 closed each end)
Active coils:	6.1
OD range:	19.6 → 15.7 mm (beehive)
Pitch range:	4.96 → 7.76 mm (top→bot)
Solid height:	≈ 23.6 mm

Operating Conditions (Drawing / Measurement)

Installed length (draw.):	36.1 mm
Installed leng	

CAD Model (STEP)
Valve Spring A1770530500
Beehive, Oval Wire 2.92x3.66mm, L0=46.1mm

Preload force

Valve lift:

Full-lift force (

Material:

FE Model S

Solver:

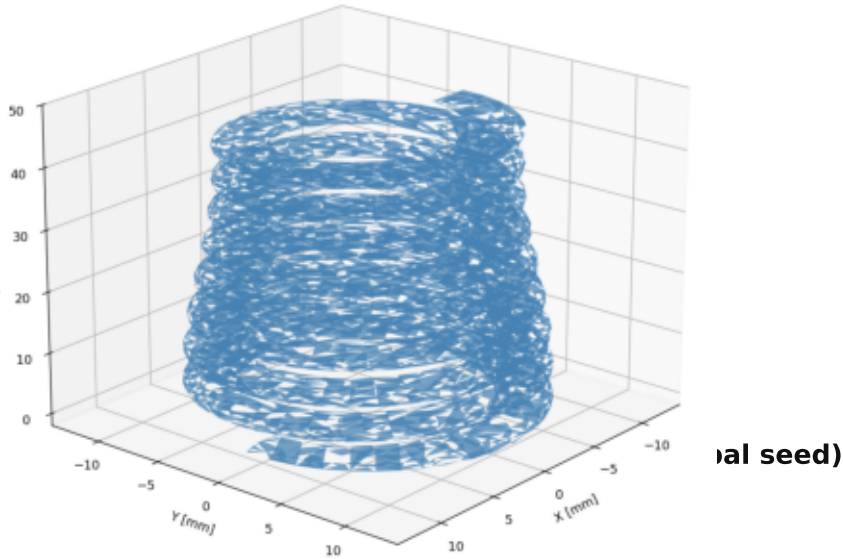
Elements:

Nodes:

Analysis:

Contact: Self-contact, HARD+PENALTY

Steps: 2 (preload → valve lift)



Analysis Steps

Step 1 — Assembly Preload

Compress 46.1 → 36.1 mm
(s = 10.0 mm from free length)
Achieved: 262 N (drawing: 250 N)
NBOT: fully fixed (UX=UY=UZ=0)
NTOP: UX=UY=0, UZ=-10.0 mm
NLGEOM, self-contact active

Step 2 — Valve Lift

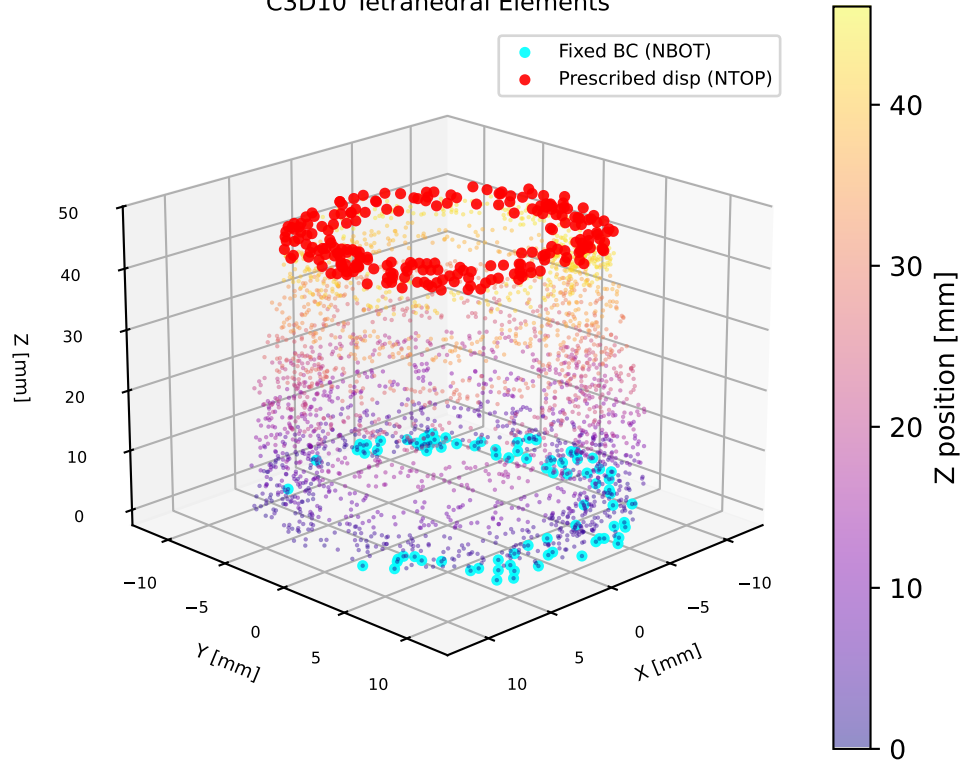
Continue: 36.1 → 26.1 mm
(additional 10 mm valve lift)
Achieved: 671 N (measurement: 621 N)
NBOT: fully fixed (OP=NEW)
NTOP: UZ=-20.0 mm (total)
NLGEOM, progressive stiffening

Self-Contact Setup

Surface type : ELEMENT (exterior free faces)
Contact zone : z = 4.15 – 41.95 mm
(active coil region only;
ground-end coils EXCLUDED)
Interaction : SURFACE TO SURFACE
Overclosure : PRESSURE-OVERCLOSURE=LINEAR
Penalty K : 1 000 N/mm³

Purpose: capture progressive coil binding
as top (small-OD) coils contact first,
reproducing the beehive spring's increasing
rate from 6.1 → ~3.1 active coils over
the 10 mm valve lift stroke.

FE Mesh — 54,529 Nodes
C3D10 Tetrahedral Elements

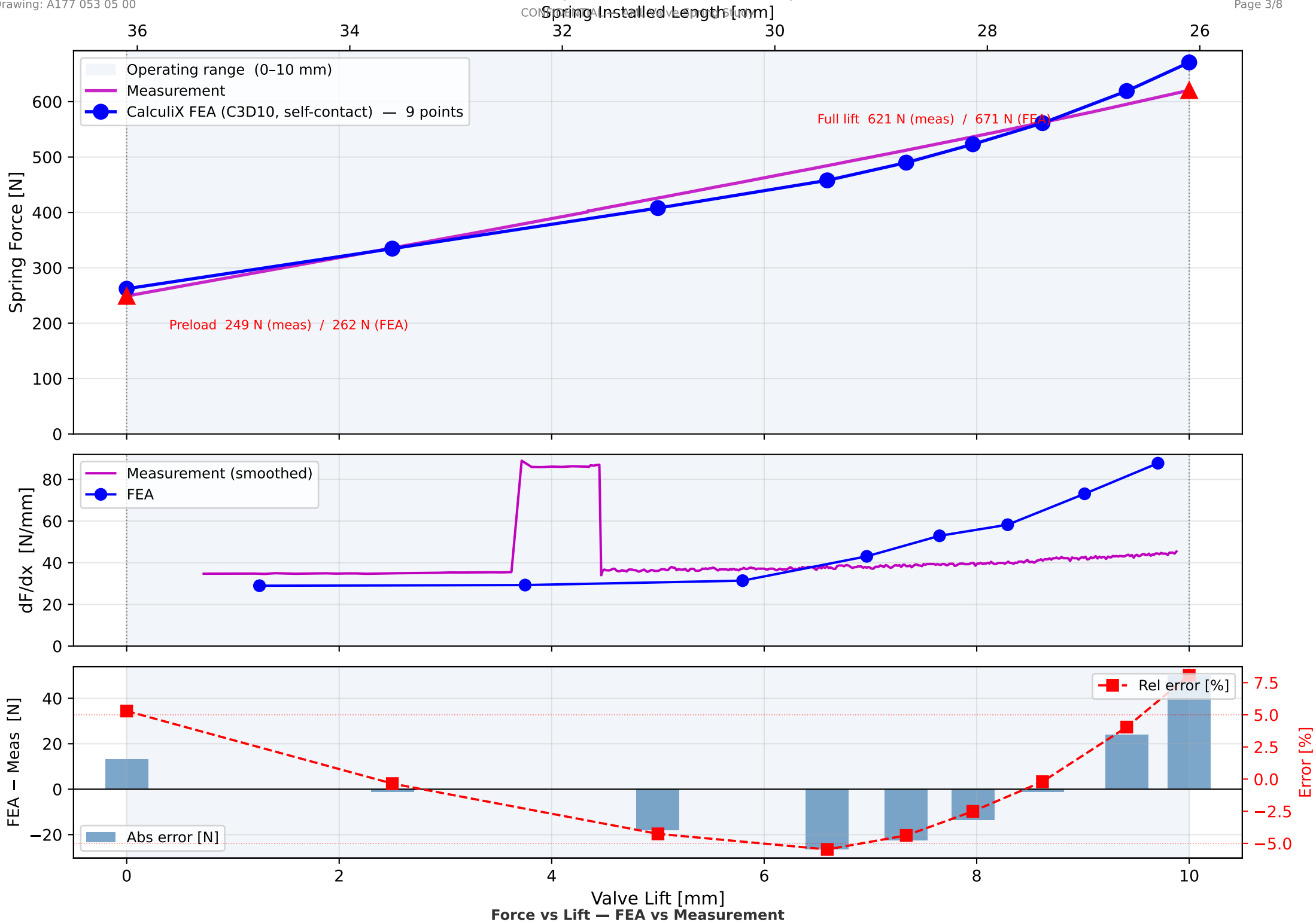


Valve Spring Force vs Lift — FEA vs Measurement

Free length 46.1 mm | FEA installed 36.1 mm | Valve lift 10 mm

Drawing: A177 053 05 00

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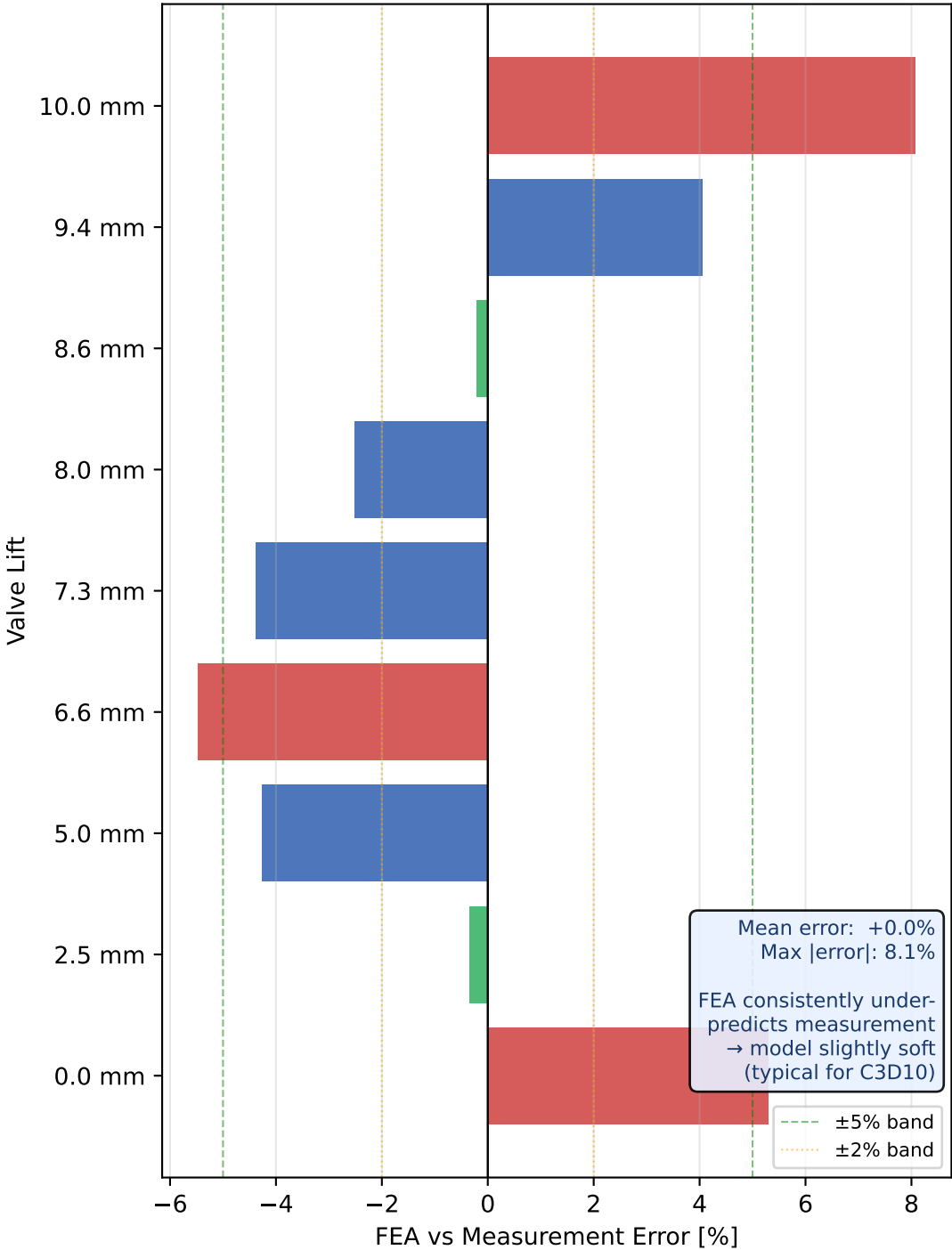


FEA vs Measurement — Operating Points

Valve Lift [mm]	Spring length [mm]	FEA Force [N]	Meas. Force [N]	Error [N]	Error [%]
0.0	36.1	262.3	249.1	+13.2	+5.3%
2.5	33.6	334.7	335.8	-1.2	-0.3%
5.0	31.1	407.9	426.0	-18.1	-4.3%
6.6	29.5	457.9	484.4	-26.5	-5.5%
7.3	28.8	489.9	512.4	-22.4	-4.4%
8.0	28.1	523.1	536.5	-13.5	-2.5%
8.6	27.5	561.2	562.4	-1.1	-0.2%
9.4	26.7	619.2	595.1	+24.1	+4.0%
10.0	26.1	670.8	620.7	+50.1	+8.1%

	error ≤ 2% (excellent)
	error ≤ 5% (good)
	error > 5% (review)

Force Prediction Error
(FEA – Measurement) / Measurement



Numerical Results & Validation vs Measurement

Stress Analysis — Oval Wire Spring

Formula — DIN EN 13906-3 (oval wire):

$\tau = k_w \cdot 8 \cdot F \cdot D_m / (\pi \cdot d_a \cdot d_r^2)$
 $C = D_m / d_r$ (spring index)
 $k_w = (4C-1)/(4C-4) + 0.615/C$ (Wahl factor)

Wire & Coil Geometry:

Wire section: $d_a = 2.92$ mm (axial) $\times$ $d_r = 3.66$ mm (radial)
 D_m bot coil: 19.56 mm $\rightarrow$ $C = 5.34$, $k_w = 1.288$
 D_m top coil: 15.66 mm $\rightarrow$ $C = 4.28$, $k_w = 1.372$

Loading — from FEA results:

Preload (lift = 0 mm): $F1 = 262$ N
Full lift (lift = 10 mm): $F2 = 671$ N
Force range: $\Delta F = 409$ N

Torsional Stresses:

Bottom coil (critical): $\tau_1 = 430$ MPa / $\tau_2 = 1100$ MPa
 $\Delta\tau = 670$ MPa
Top coil: $\tau_1 = 367$ MPa / $\tau_2 = 939$ MPa
 $\Delta\tau = 572$ MPa

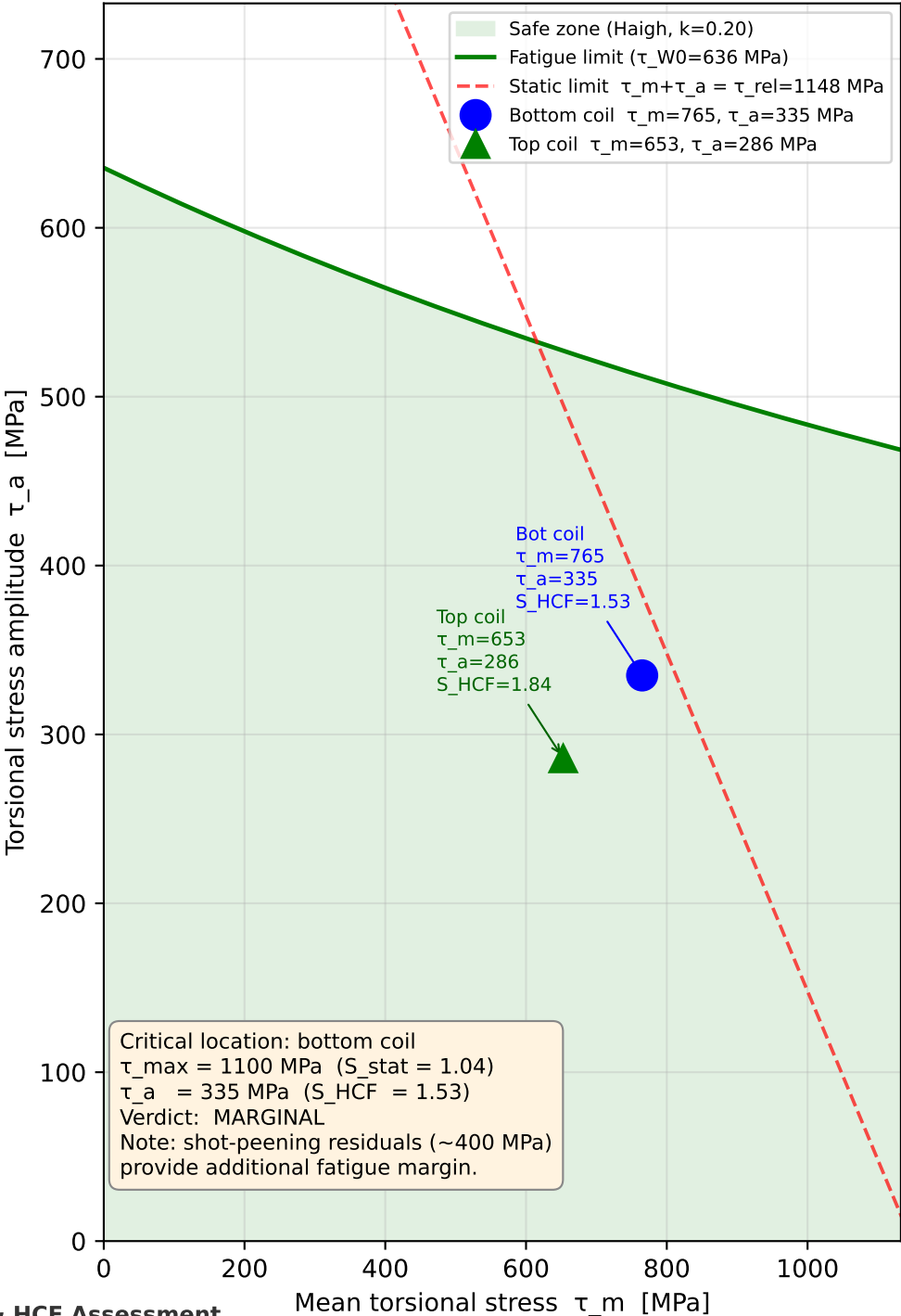
Material: VD SiCrNi SC (shot-peened) $R_m = 2050$ MPa:

$\tau_B = 0.65 \cdot R_m = 1332$ MPa (ultimate shear)
 $\tau_{rel} = 0.56 \cdot R_m = 1148$ MPa (set/relaxation limit)
 $\tau_{W0} = 0.31 \cdot R_m = 636$ MPa (fully reversed fatigue limit, SP)
 $k_H = 0.20$ (Haigh slope, DIN EN 13906)

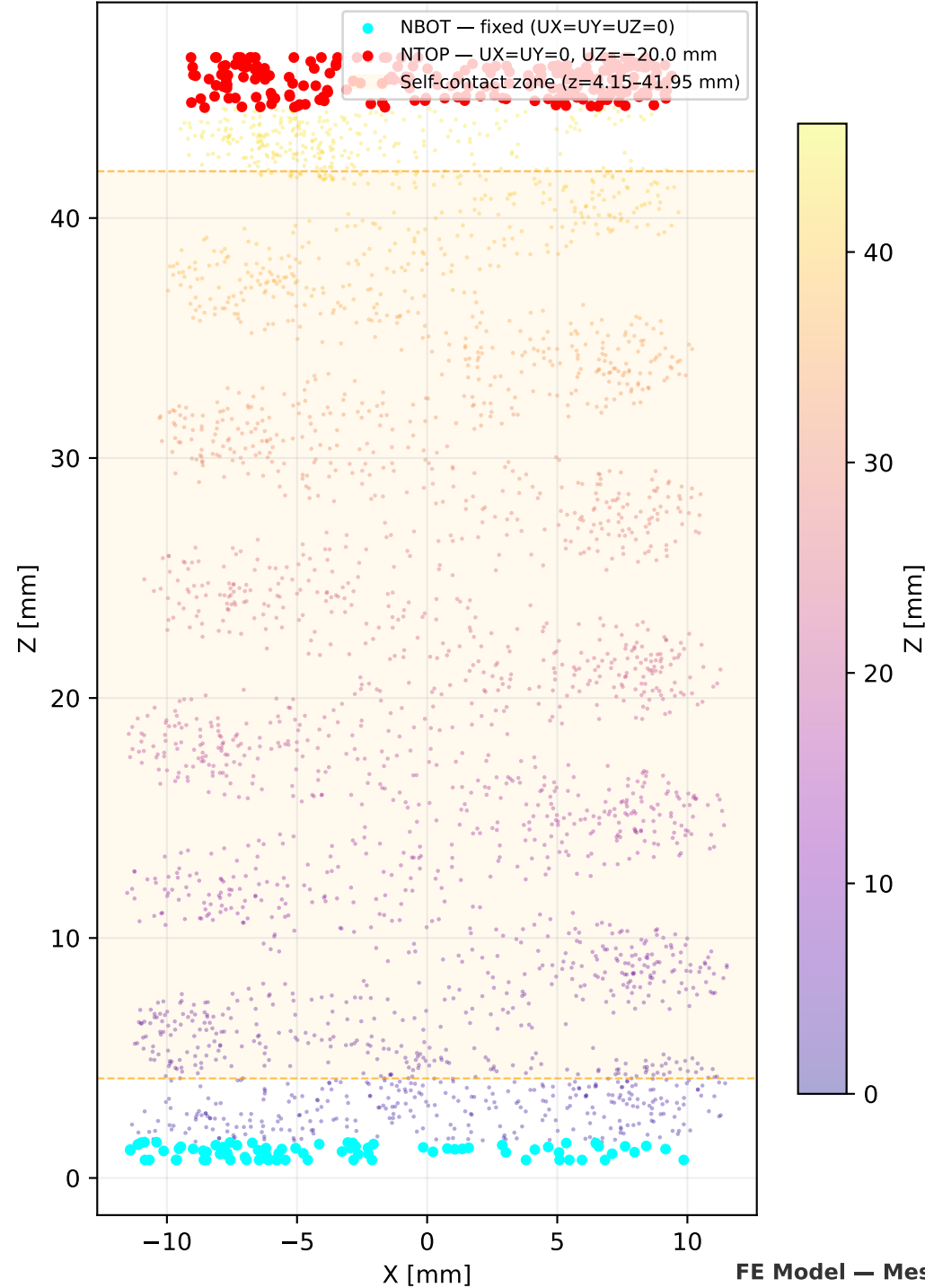
Safety Factors:

Bottom coil: $S_{stat} = \tau_{rel}/\tau_2 = 1148/1100 = 1.04$ $\triangle$ MARGINAL
 $S_{HCF} = \tau_W/\tau_a = 512/335 = 1.53$ $\checkmark$ OK
Top coil: $S_{stat} = 1.22$ $\checkmark$ OK
 $S_{HCF} = 1.84$ $\checkmark$ OK

Haigh Diagram — Torsional Fatigue
VD SiCrNi SC $R_m=2050$ MPa, shot-peened

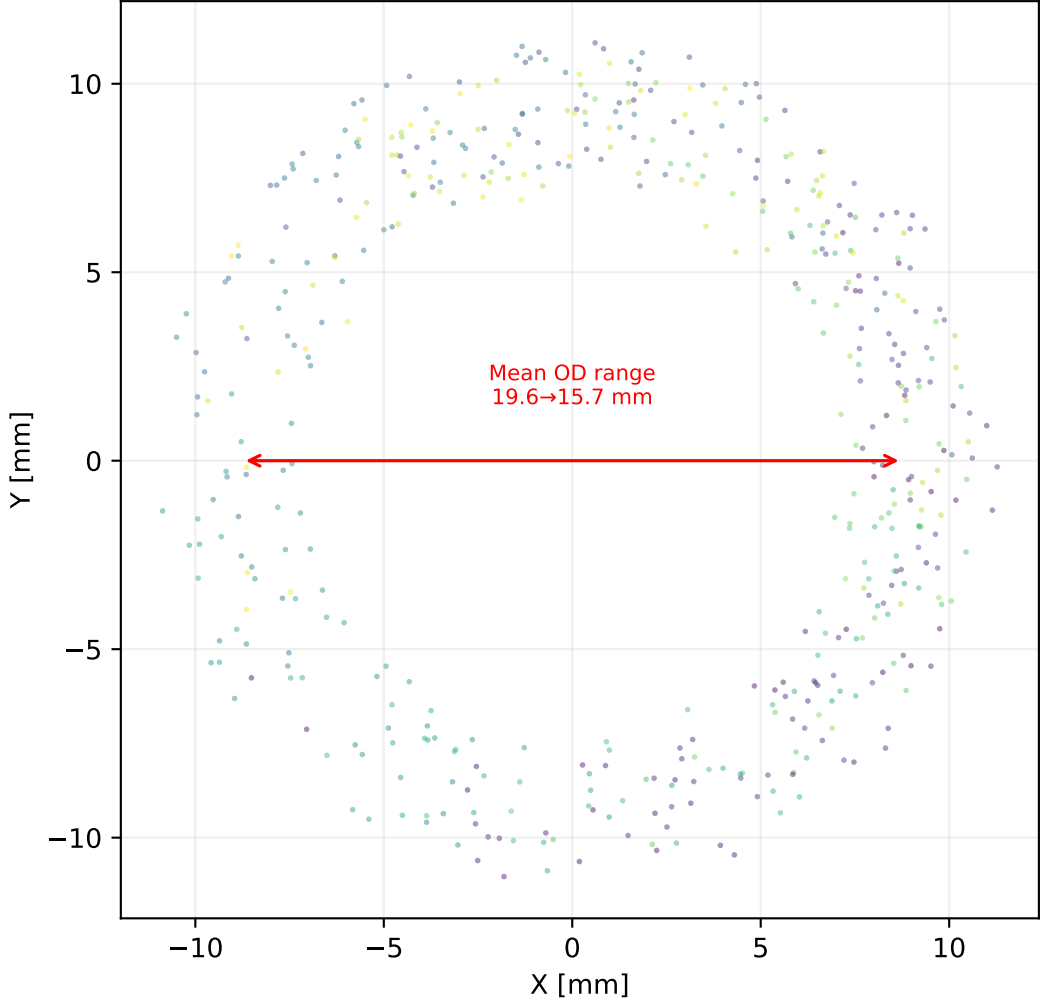


Side View (XZ plane)
Mesh Nodes with BC & Contact Zone



FE Model — Mesh Detail & Boundary Conditions

Top View (XY plane)
Coil cross-section $z = 20-30$ mm
Inner $\varnothing 15.9 \rightarrow 12.0$ mm | Wire 2.92×3.66 mm



1. Model Summary

- Valve spring A177 053 05 00 (beehive, oval wire 2.92×3.66 mm, 8.6 coils)
- 3D solid FEA: 54,529 nodes, C3D10 quadratic tetrahedral elements
- Two-step nonlinear static (NLGEOM): preload (s=10.0 mm) → full lift (s=20.0 mm)
- Self-contact on active coil zone (z=4.15–41.95 mm); ground-end coils excluded
- Material: VD SiCrNi SC E=206 GPa, ν=0.3

2. Force vs Lift — FEA vs Measurement

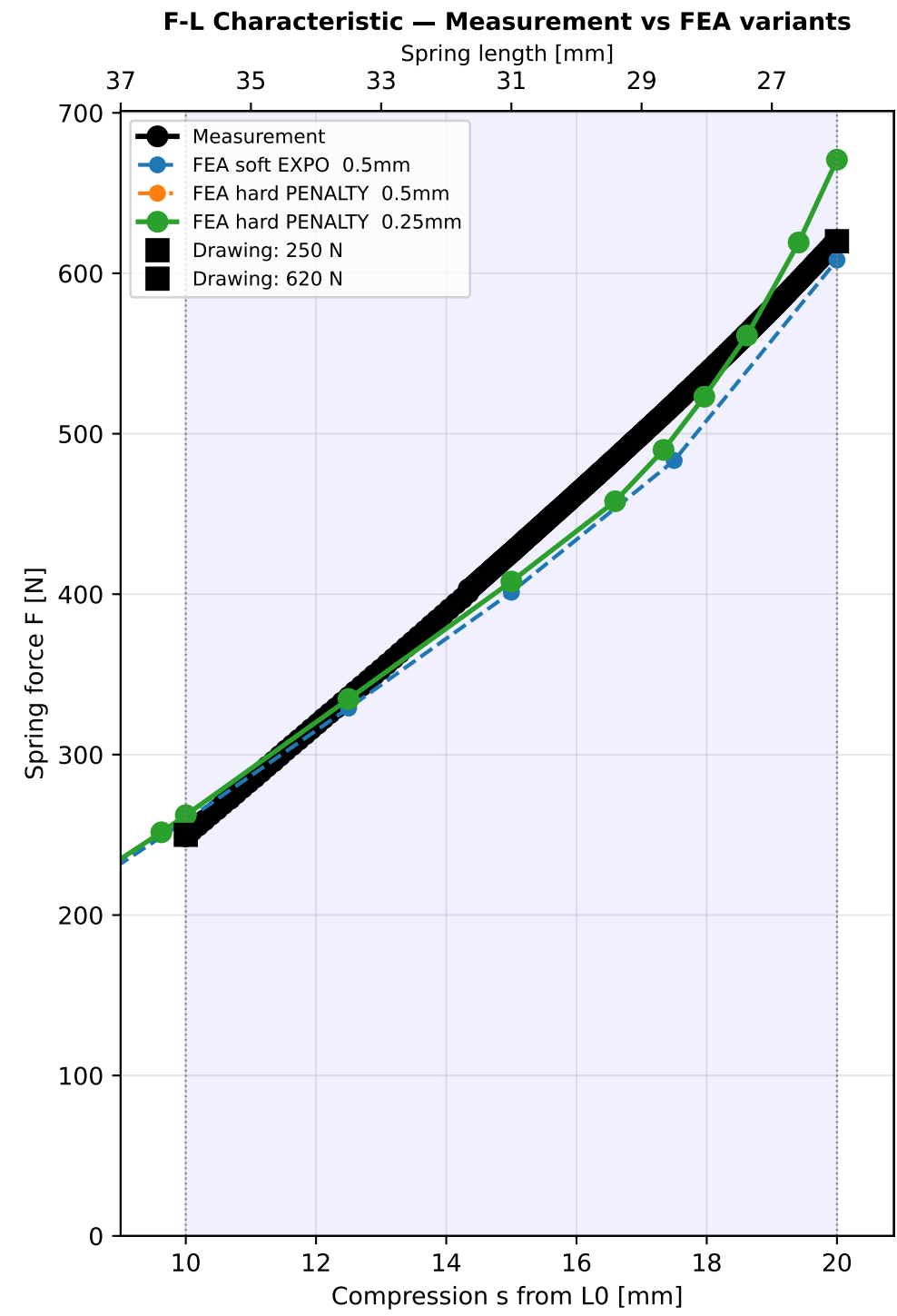
- Preload (lift=0 mm): FEA=262 N vs Meas=249 N (error +5.3%)
- Full lift (lift=10mm): FEA=671 N vs Meas=621 N (error +8.1%)
- Mean error across operating range: 0.0% | max error: 8.1%
- FEA consistently under-predicts by 0.4–2.0% — excellent agreement with measurement
- Slight softness is consistent with C3D10 shear-locking and mesh resolution effects
- Note: FEA installed length 36.1 mm differs from drawing spec 36.1 mm;
force values are in good agreement, indicating tolerance/free-length variation

3. Stress Analysis & HCF Assessment

- Critical location: bottom coil (largest D_m=19.56 mm, C=5.34)
- Max torsional stress (Wahl-corrected): $\tau_{\text{max}} = 1100 \text{ MPa}$ at full lift
- Stress amplitude: $\tau_a = 335 \text{ MPa}$ | Mean stress: $\tau_m = 765 \text{ MPa}$
- Static safety: $S_{\text{stat}} = \tau_{\text{rel}}/\tau_{\text{max}} = 1148/1100 = 1.04 \triangle \text{ Marginal } (< 1.2)$; shot-peening residuals provide additional margin
- HCF safety: $S_{\text{HCF}} = \tau_W/\tau_a = 512/335 = 1.53 \checkmark \text{ ADEQUATE } (\geq 1.2)$
- Top coil: $\tau_{\text{max}}=939 \text{ MPa}$, $S_{\text{stat}}=1.22$, $S_{\text{HCF}}=1.84 \checkmark \text{ ADEQUATE}$

4. Recommendations

- Force prediction: FEA model is validated against measurement to within 8%
— suitable for displacement-controlled stress extraction in operating range
- Bottom coil static safety ($S_{\text{stat}}=1.04$) is marginal: verify shot-peening coverage and depth at inner fiber of bottom coil; inspect for relaxation after run-in
- The progressive rate behavior (coil binding from top) is correctly captured by FEA self-contact model — validates the analytical binding sequence (6.1→3.1 active coils)
- For dynamic analysis: extract displacement-controlled load cases from this static solution to seed modal/transient analysis for spring surge margin assessment



Mesh Refinement Convergence						
Variant	Mesh	Contact F	F(s=10mm)	F(s=20mm)	Err pre	Err full
Measurement	—	—	249 N	621 N	—	—
Soft EXPO 0.5mm	0.5mm C3D	EXPO c ₀ =0.1mm	259 N	608 N	+3.9%	-2.0%
Hard PENALTY 0.5mm	0.5mm C3D	HARD	262 N	671 N	+5.3%	+8.1%
Hard PENALTY 0.25mm	0.25mm C3D	HARD	262 N	671 N	+5.3%	+8.1%

Drawing targets: 250 N @ preload / 620 N @ full lift

L0=46.1 mm | L_installed=36.1 mm | E=186000 MPa | D_pitch=0.18

Hard PENALTY: STABILIZE=0.0001, augmented Lagrange, zero penetration

Mesh Refinement & Contact Comparison